Battery voltage divider for measuring voltage

R4 = 470
R5 = 680

Voltage Division Equation
Vadc = Vbatt * (R5 / (R4 + R5))

Voltage Vbatt --> Vadc ratio
ADC_Division = Vadc / Vbatt = (R5 / (R4 + R5))
R5 / (R4 + R5) = 0.5913043478260087

Voltage Vadc --> Vbatt ratio
ADC_Multiplier = Vadc / Vbatt = (R4 + R5) / R5
(R4 + R5) / R5 = 1.6911764705882353

C1 used for getting more stable and accurate readings

firmware/variants/nrf52840/diy/nrf52_promicro_diy_tcxo/variant.h
#define VBAT_DIVIDER_COMP (1.73)
#define ADC_MULTIPLIER VBAT_DIVIDER_COMP

+BATT, Vbatt, R4 680K, Batt%, Vadc, C1 100nF, R5 1.0M, GND

HT-RA62 / RA-01SH

U2 HT-RA62
ANT, GND, 3V3, RST, TXEN, DIO1, DIO2, DIO3, NSS, MOSI, MISO, SCK, RXEN, BUSY, CS
HT-RA62
Heltec

Pins for connecting an I2C OLED screen

J5 I2C Oled screen: SCL, SDA
OLED1 I2C OLED Module: I2C OLED SSD1306 0.96inch

Pads for connecting encoder

ENC_A HW GPIO: P1.01 PlatformIO GPIO: 33
ENC_B HW GPIO: P1.02 PlatformIO GPIO: 34
ENC_PUSH HW GPIO: P1.07 PlatformIO GPIO: 39

R11 10k, R10 10k, R9 10k
J16 Encoder Pads
Pick any EC11 encoder that fits your enclosure

J50 SM04B-SRSS-TB JST Sales America Inc. JST-SH-4

SW3 RotaryEncoder_Switch
ACZ11BR1E-15KQA1-12C Same Sky (Formerly CUI Devices)

Onboard buttons RST

SW1 Reset, RST_BTN, XUNPU TS-1088R-02026
Onboard buttons, RST for resetting

Onboard buttons BTN

SW2 SW_Push, BTN, XUNPU TS-1088R-02026
Onboard buttons, BTN is programmable, connected to D6 (P1.00)

Nice!Nano nrf52840 microcontroller (or promicro)

U3 nice!nano v2
GND, P0.06, P0.08, GND, GND, P0.17, P0.20, P0.22, P0.24, P1.00, P0.11, P1.04, P1.06, P1.01, P1.02, P1.07
BAT+, BAT+, GND, RST, 3.3v, P0.31, P0.29, P0.02, P1.15, P1.13, P1.11, P0.10, P0.09
FET3, FET2, RXEN, GPSTX, GPSRX, FET1, BTN, SCL, SDA, EXTRA
Batt%, BUSY, MISO, MOSI, CS, SCK, DIO1, RST
RST_BTN
ENC_PUSH, ENC_B, ENC_A

nice!nano v2
Nice!

Mosfets used for switching high current external perpherial

AO3400A Alpha & Omega Semiconductor Inc.

FET1_GPSEN — HW GPIO: P0.24 PlatformIO GPIO: 24
FET1 is also connected to the GPS pads' Ground.
Q1 AO3400A, R1 10k, GPS_GND, J6 FET1 Drain, J11 FET1 3,3
Dev Note: This is for turning on/off the power to a gps module to minimise power usage

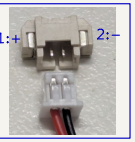
FET2_VIB — HW GPIO: P0.08 PlatformIO GPIO: 8
Alternatively can use SI2312 (More expensive, Higher current)
Q2 AO3400A, R2 10k, VIBRATION, M1 ERM Motor, J7 FET2 Drain, J12 FET2 3,3
316040001 Seeed Technology Co., Ltd

FET3_BUZ — HW GPIO: P0.06 PlatformIO GPIO: 6
Each MOSFET is part of a 3v3/ground pair of pads, with the FET controlling the ground.
Q3 AO3400A, R3 10k, BUZZER, J8 FET3 Drain, J13 FET3 3,3
HYDZ HYG-5020A, Buzzer 5020 BZ1

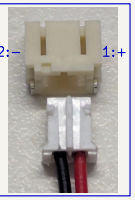
Battery connector

J9 Molex PicoBlade 0532610271 CJT(Changjiang Connectors) A1251WR-S-2P
PWR_FLAG +BATT PWR_FLAG -BATT PWR_FLAG
J46 Conn_01x02_Socket -BATT +BATT

Molex Picoblade 2-Pin
- MPC: Molex, MPN: 0532610271
- MPC: CJT (Changjiang Connectors), MPN: A1251WR-S-2P
- Pitch: 1.25mm
- Used By: HelTec LoRa 32
- This polarity is also common for lipos in Aliexpress



JST PH 2-Pin
- MPN:S2B-PH-SM4-TB
- Pitch: 2.00mm
- Used By: RAK19007 WisBlock
- This polarity is also common for lipos in Aliexpress



BMS for overdischarge protection

U1 XB8089D0
JP1 Jumper_2_Open
Short JP1 to bypass battery protection if XB8089D0 is not populated
C2 100nF, R6 1k, PWR_FLAG, -BATT, +BATT
XySemi XB8089D0
The N!ce Nano already has overvoltage protection, but does not prevent overdischarge during use. This BMS ensures that the device shuts off if the battery voltage drops below 2.9v

GPS Module

J14 GPS Pins: GPS_GND, GPSRX, GPSTX
Solder Pad Option
J51 SM04B-SRSS-TB JST Sales America Inc. JST-SH-4
Socket Option
e.g. Buy either ATGM336H or GY-GPS6MV2 modules
Dev Note: Intent is to cut an JST-SH-4 (or qwiic) cable in half and solder one to an EC11 rotary encoder and the other to the GPS module.

I2C Port (For extra sensors etc...)

J41 GND, J40 3.3V, J39 SDA, J37 SCL
J38 JST-SH-4 SM04B-SRSS-TB JST Sales America Inc.
The JST-SH buddies
https://hackaday.com/2022/05/04/the-connector-zoo-i2c-ecosystems

EXTRA GPIO

EXTRA — HW GPIO: P1.06 PlatformIO GPIO: 38
J1 EXTRA, J45 GND

Development Source: https://github.com/mofosyne/roTec_pcb
Fork Of FakeTec: https://github.com/jtljac/fakeTec_pcb
Fork Of FakeTec: https://github.com/gargomoma/fakeTec_pcb/

Sheet: /
File: fakeTecv5.kicad_sch

Title: roTec Rev5.2.2

Size: A4	Date: 2025-12-26	**Rev: 5.2.2**
KiCad E.D.A. 9.0.4		Id: 1/1